

A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision

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Abstract

Most quant stacks start from *strategy*. Public LLMs (GPT-, DeepSeek-, GLM-, and Gemini-class agents) that act in cryptocurrency markets fail earlier: they lack a complete, honest, auditable *market cognition base*. We present an anonymized *financial world-model runtime* that compiles asynchronous multi-band evidence into a point-in-time (PIT) world bundle. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{\text{AI}}$; completeness, honesty (B, U, H), and WMI/ACWMI support a typed abstention contract; evidence IDs make actions auditable. Four live public LLMs (temperature 0) on a real crypto PIT panel show a three-arm behavior ladder: Compiled thin-world abstain 1.0; Ungated (same content, no hard flag) mean 0.74; Raw 0.04–0.75. Disclosure helps; the typed boolean enforces full cross-vendor refusal. Economic CE is a secondary nonempty-content probe—not a strategy Holy Grail. We detail the runtime architecture (collectors, vintage store, BandPIT clock, bundle schema, LLM adapter) as a systems contribution for ICAIF.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, market cognition, systems

ACM Reference Format:

Anonymous Author(s). 2026. A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision. In *Proceedings of the 7th ACM International Conference on AI in Finance (ICAIF '26)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/XXXXXXX.XXXXXXX>

1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [11, 13]. Generative world models [5, 6, 9] and tool-using finance agents [18–20] typically assume a usable observation stream. Crypto agents often do not have one.

Thesis. We build a market *world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe cognition bundle that GPT / DeepSeek / GLM / Gemini-class models can call. Strategy metrics are secondary probes that the world has content—not the

product objective. In slogan form: *quant from strategy; this runtime from understanding*.

What we claim (and do not). RQ1 tests whether a typed world interface can *enforce* world-conditional abstention across live public LLMs—a runtime contract, not a claim that models spontaneously discover thinness from raw ticks. RQ2 tests whether compiled band fields are economically nonempty versus same-input momentum. We do not claim unconditional trading alpha or a Dreamer-style generative simulator.

Contributions.

- (1) **Cognition semantics.** Epistemic observations, compilation Π_t , a lag reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
- (2) **Anonymized runtime architecture.** Vintage-aware collectors, BandPIT previous-close clock, quality-tagged bundles, availability shocks O_t , OpenAI-compatible multi-vendor adapter, and a frozen Compiled / Ungated / Raw consumer protocol.
- (3) **Live three-arm validation.** Four live public LLMs: Compiled enforces thin-world abstain 1.0; Ungated soft judgment averages 0.74; Raw often trades. A secondary content probe shows vintaged band fields are economically nonempty.

2 Related Work

World models in AI. World models learn compact states and often dynamics [5, 6, 9]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing systems layer generative WMs usually assume—not a Dreamer/JEPA simulator.

LLM agents in finance. ChatGPT return prediction [12], open financial LLMs [17], memory-augmented trading agents [10, 19], and tool-augmented multimodal agents [20] advance agent skill. Agentic tool use [18] still assumes aligned inputs. We evaluate within-model Compiled vs. Ungated vs. Raw under a frozen schema so the estimand is the *world interface*, not a new memory module or prompt trick.

Selective prediction and trustworthy AI. Abstention can be optimal under noise [2, 3]. We tie refusal to world quality via an explicit runtime contract, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [1] and crypto microstructure [11, 13] motivate multi-band compilation. Bootstrap discipline for secondary probes follows [7, 15, 16]. Asset-pricing ML [4, 8, 14] is adjacent: we validate nonempty compiled content for agents, not priced factors.

3 Market World-Model Semantics

DEFINITION 1 (EPISTEMIC OBSERVATION). $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$ ³ *value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.*

ICAIF '26, Milan, Italy

2026. ACM ISBN 978-1-4503-XXXX-X/26/11

<https://doi.org/10.1145/XXXXXXX.XXXXXXX>

DEFINITION 2 (FINANCIAL WORLD STATE). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (e.g. WMI/ACWMI) gating abstention; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.

DEFINITION 3 (COMPILATION). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled contract exposes `should_ai_abstain` and `thin_world` as first-class fields.

ASSUMPTION 1 (BOUNDED LAG AND INCREMENTS). Band j has max observation lag Δ_j ; latent state increments satisfy $\|S_u - S_{u-1}\| \leq \delta$.

PROPOSITION 1 (LAG RECONSTRUCTION BOUND). Under Assumption 1,

$$\|\tilde{S}_t - S_t\| \leq C_{\text{del}}\delta + C_{\text{noi}}\epsilon_t + C_{\text{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 1 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

PROPOSITION 2 (COMPILATION $\neq$ FEATURE DUMP). Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty/stability.

PROPOSITION 3 (WORLD-CONDITIONAL ABSTENTION). If non-abstain actions exceed world-dependent cost $c_{\text{abs}}(W_t)$, Bayes action is abstain; with costs decreasing in WMI, abstention is a lower set in world quality [2].

PROPOSITION 4 (LOBO CONTENT VS. GATING). Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.

Reading for systems reviewers. Props. 2–3 justify why the runtime exports gates and abstain flags rather than only tilts. Prop. 4 justifies the secondary economic probe: if deleting a durable band does not change actions, the bundle is decorative.

REMARK 1 (NOT A GENERATIVE WM). We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler + abstention runtime.

4 Runtime Architecture

We describe the anonymized prototype as a layered systems object (Figure 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

(1) **Data layer (collectors).** Exchange, macro, alternative, and (when available) news/on-chain/options collectors write vintage-aware history with observation timestamps and macro `available_at` fields.

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t , WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw sampling, transcripts, tables

- (2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI/ACWMI, paper world snapshots).
- (3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty/missingness gates and band assembly implement Π_t ; world-quality indices and availability shocks O_t are first-class.
- (4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
- (5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

4.2 PIT previous-close clock

For calendar day t , the decision information set is reconstructed at $(t-1)$ 23:59. The payoff is the same-calendar-day close-to-close return r_t . This removes same-day look-ahead in band statuses and vintaged macro/alternative content. Bundle field `decision_asof` records the clock; `timing_protocol` is fixed to `decision_at_prev_close` for all live runs.

4.3 World bundle contract

Table 2 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: only 2/8 bands ready, WMI low, `should_ai_abstain=true`, and `evidence_ids` binding actions to disclosed bands/tilts. Ungated removes the hard boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only `mom5` (plus a non-informative noise bit).

Listing 1: Compact Compiled bundle example (OOS thin day).

```
{
  "date": "2026-05-11", "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {"n_ready": 2, "n_missing": 6,
    "ready_share": 0.25},
  "world_model_index": {"wmi": 0.015, "acwmi": 0.30,
    "should_ai_abstain": true, "thin_world": true},
  "macro_tilt": 1.0, "alt_tilt": -1.0,
  "detected_regime": "crisis",
  "audit": {"evidence_ids": [
    "band:exchange:missing", "band:macro:ready",
    "band:alternative:ready", "tilt:macro:1.0"]}
}
```

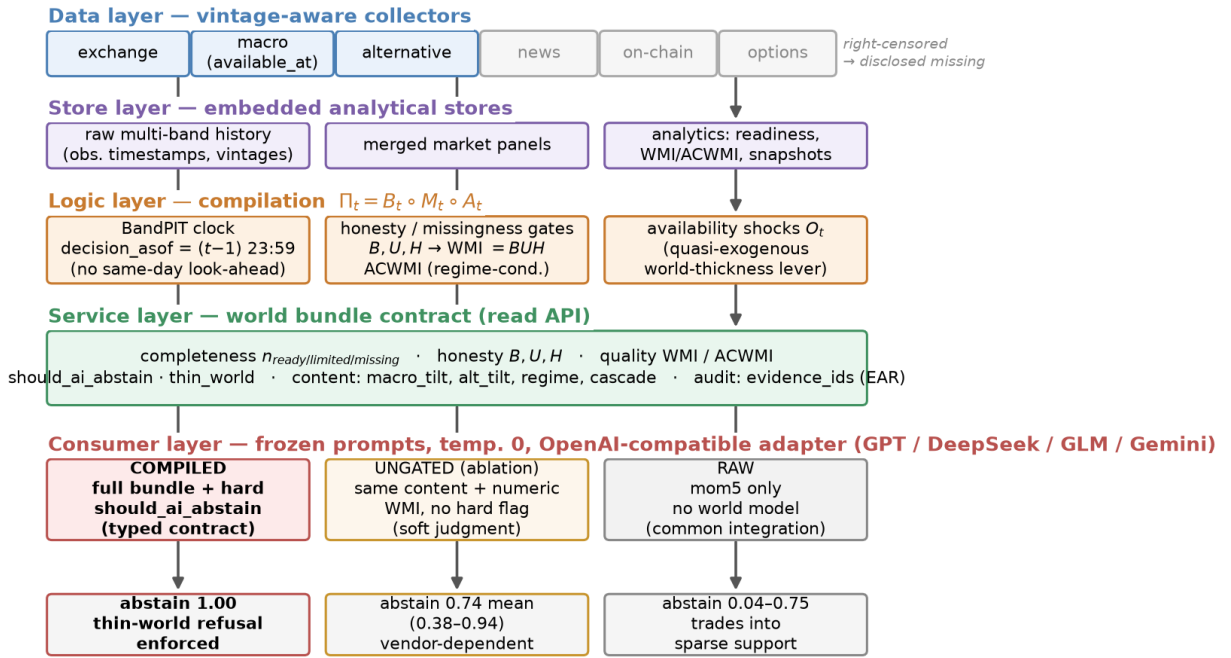


Figure 1: Anonymized runtime architecture. Five layers map raw multi-band evidence to a typed world-bundle contract; the consumer layer runs the frozen three-arm protocol (Compiled / Ungated / Raw), whose live abstention outcomes are shown at the bottom.

Table 2: World bundle field groups (cognition base).

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	$n_{ready/limited/missing}$, band statuses
Honesty	B, U, H , main-view / stale gates
Quality	WMI, ACWMI, should_ai_abstain, thin_world
Content	macro_tilt, alt_tilt, regime, cascade
Audit	evidence_ids, EAR required

4.4 Three information arms and LLM adapter

- **Compiled (product contract):** full bundle; frozen prompt requires abstain when should_ai_abstain is true.
- **Ungated (ablation):** same content/completeness/numeric WMI; no hard flag; soft prompt asks the model to judge thinness.
- **Raw (without world model):** mom5 only; no quality index or abstention guidance.

Actions $\in \{\text{bullish, bearish, neutral, abstain}\}$. Understanding metrics prioritize thin-world abstain rate and EAR proxy over CE. Durable archive bands: {exchange, macro, alternative}; news/on-chain/options are right-censored and disclosed as missing (Figure 2). The adapter speaks OpenAI-compatible chat completions; model IDs are passed through; credentials stay in environment variables (never in artifacts).

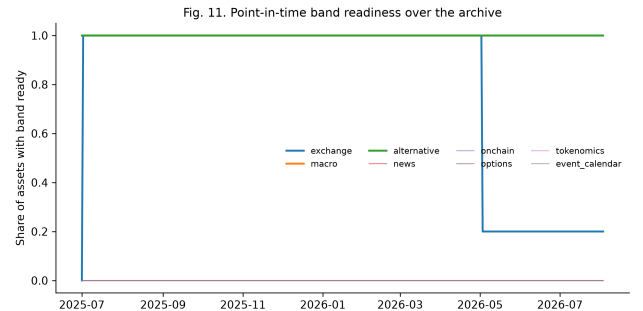


Figure 2: PIT readiness: missing bands are first-class in the bundle.

4.5 Frozen prompt contracts

Listings 2–3 excerpt the frozen Compiled and Ungated contracts (temperature 0; action schema identical). The only intentional difference is hard vs. soft abstention. Raw prompts omit world-quality language entirely.

Listing 2: Compiled contract (excerpt).

```
Decision rules:
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
4. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft contract (excerpt).

```

Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.

```

Not “just instruction following.” Compiled abstention is partly prompt obedience—and that is the *product requirement*: production agents need an enforceable refuse interface when $q(W_t)$ is low (Prop. 3). Ungated measures soft judgment from disclosure alone; Raw measures the common “price signal only” integration. Reliable Compiled refusal is the SLA; spontaneous Ungated refusal is a bonus.

4.6 Service surface and reproducibility

The service layer exposes read-only objects used by the consumer harness: (i) quality-tagged world bundles; (ii) band readiness / WMI-ACWMI snapshots; (iii) availability-shock queries O_t ; (iv) health metadata disclosing whether paper engines or production proxies hydrate ACWMI inputs. Collectors and the logic pipeline can run as supervised jobs; the API reads stores live, so newly compiled rows appear without process bounce. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and an OpenAI-compatible base URL are configuration, not estimands. Credentials never enter committed artifacts. Upon acceptance we will release an anonymized repository with the consumer harness, frozen prompts, and table-reproduction scripts.

5 Experiments**5.1 Setup**

PIT panel ~399 days (≈ 13 months), 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days. Live RQ1: stratified OOS sample, temperature 0, frozen prompts, OpenAI-compatible gateway. Models: gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite. Compiled/Raw: 100 asset-days each; Ungated: 50 asset-days from the *same sampling seed* (subset, not a disjoint regime). Offline mocks reproduce the protocol on full OOS without keys. Scarce/thin labels bind on OOS by design: primary test is honesty under sparse support, not a dense-world trading horse-race.

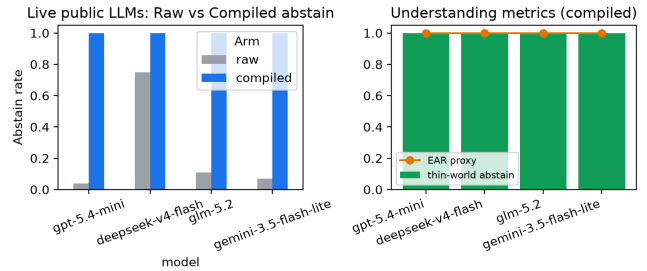
5.2 RQ1 (primary): Three-arm live behavior

Table 3 is the headline result. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world fields on 93–99% of days. Under Ungated, mean thin-abs. falls to 0.74 (range 0.38–0.94): three models still refuse often, but gemini-3.5-flash-lite drops to 0.38 and trades. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 4). Mean ΔCE (Compiled–Raw) = +1.07 is reported last: Compiled CE is zero under full abstention, so the gap mainly reflects Raw overtrading—not a Holy Grail.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only and noisy mocks ignore thin-world flags (thin-abs. 0)—the Raw failure mode in miniature. Offline results are protocol checks; live vendors are the product claim.

Table 3: Live three-arm ladder. Compiled/Raw: 100 OOS days; Ungated: 50-day same-seed subset. ΔCE secondary.

Model	Thin C	Thin U	Abs. R	ΔCE
gpt-5.4-mini	1.00	0.80	0.04	+0.73
deepseek-v4-flash	1.00	0.82	0.75	+1.22
glm-5.2	1.00	0.94	0.11	+1.04
gemini-3.5-flash-lite	1.00	0.38	0.07	+1.30
Mean	1.00	0.74	—	+1.07

**Figure 3: Compiled vs. Raw understanding metrics (live). Ungated numbers in Table 3.****Table 4: Raw-arm action mix (100 days). Compiled is 100% abstain.**

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 5: Secondary content probe (not the LLM product claim).

Policy / contrast	Sharpe	CE
Momentum always	0.101	−0.202
Thick ungated (content)	0.767	0.132
Mechanism – Momentum	—	0.334 ($p=0.034$)

5.3 RQ2: Nonempty compiled content

A transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt. Pre-specified OOS contrast mechanism – momentum: $\Delta CE=0.334$ ($p=0.034$; stationary $p=0.022$). LOBO (Prop. 4): deleting macro or alternative collapses to momentum (content share 1.0 under the ungated rule). Relative gaps survive 10–25bps costs; absolute CE is cost-fragile. A 2017–2026 audit without vintaged archives finds no hidden return-rule alpha.

5.4 Discussion and threats

The live ladder $Abs_C = 1.0 > Abs_U \approx 0.74 > Abs_R \in [0.04, 0.75]$ is the product evidence: a cognition base that marks the world thin induces vendor-stable refusal; disclosure alone is incomplete; raw feeds over-act. Cross-vendor heterogeneity under Ungated (GLM 0.94 vs. Gemini 0.38) is exactly why a typed contract is valuable in production: soft judgment is model-dependent, while Compiled refusal is stable. RQ2 shows band fields are not decorative. Claim: *understanding first, strategy second*.

Relative to FinMem / FinAgent-style systems [19, 20], we do not propose a new memory or tool router. We propose the observation compiler those agents implicitly need when the market feed is asynchronous and quality-heterogeneous. Relative to return-prediction LLM studies [12], we score refusal and auditability before PnL.

Threats to validity. (1) *Prompt contract*—Compiled refusal is partly instructed; Ungated is the soft contrast. (2) *Thin archive*—scarce rate ≈ 1 on OOS stresses honesty, not dense calibration. (3) *Base-lines*—Raw matches common thin integrations; Ungated is the same-content control. (4) *Sample sizes*—100/50 live days; harness supports full ~ 2000 -row OOS sweeps. (5) *Vendor routing*—gateway and model IDs vary; prompts/schema are frozen for within-model contrasts. (6) *Task scope*—the consumer action space is a coarse directional schema; richer analysis tasks (risk narratives, user warnings) are compatible with the same bundle and left to follow-on measurement.

6 Design Rationale for ICAIF Systems Readers

Three implementation choices are deliberate.

Export gates, not only features. Silent feature stores hide staleness. By placing completeness, honesty, and `should_ai_abstain` in the bundle, the runtime makes refusal a first-class machine-readable event rather than a free-text apology.

Previous-close PIT over wall-clock streaming. Intraday streaming demos look impressive but confound look-ahead. The previous-close clock is conservative and auditable: every live decision cites a `decision_asof` stamp that a reviewer can recompute from the panel.

Three arms instead of one leaderboard. A single Compiled-vs-holdout Sharpe invites Holy-Grail readings. Compiled / Ungated / Raw separates (i) typed enforcement, (ii) soft judgment from disclosure, and (iii) no world model—matching how teams actually integrate LLMs.

7 Limitations

We do not claim generative dynamics or unconditional trading alpha. News/on-chain/options history is right-censored and disclosed as missing. Dense-world selective-prediction horse-races require thicker archives. Live samples are budget-constrained (100/50 asset-days); the harness already supports full OOS replication. Camera-ready will restore author identity and point to an anonymized repository deferred during review.

8 Conclusion

Public LLMs need a market to *understand* before they decide. We contribute cognition semantics and an anonymized layered runtime—collectors, vintage store, BandPIT compilation, bundle contract, and LLM adapter—that turns asynchronous crypto evidence into a complete, honest, auditable world state. Live public LLMs exhibit a three-arm ladder: Compiled enforces thin-world refusal, Ungated partially recovers it from disclosure alone, and Raw often trades into sparse support. Content probes confirm the compiled state is nonempty. Strategy remains downstream of understanding.

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